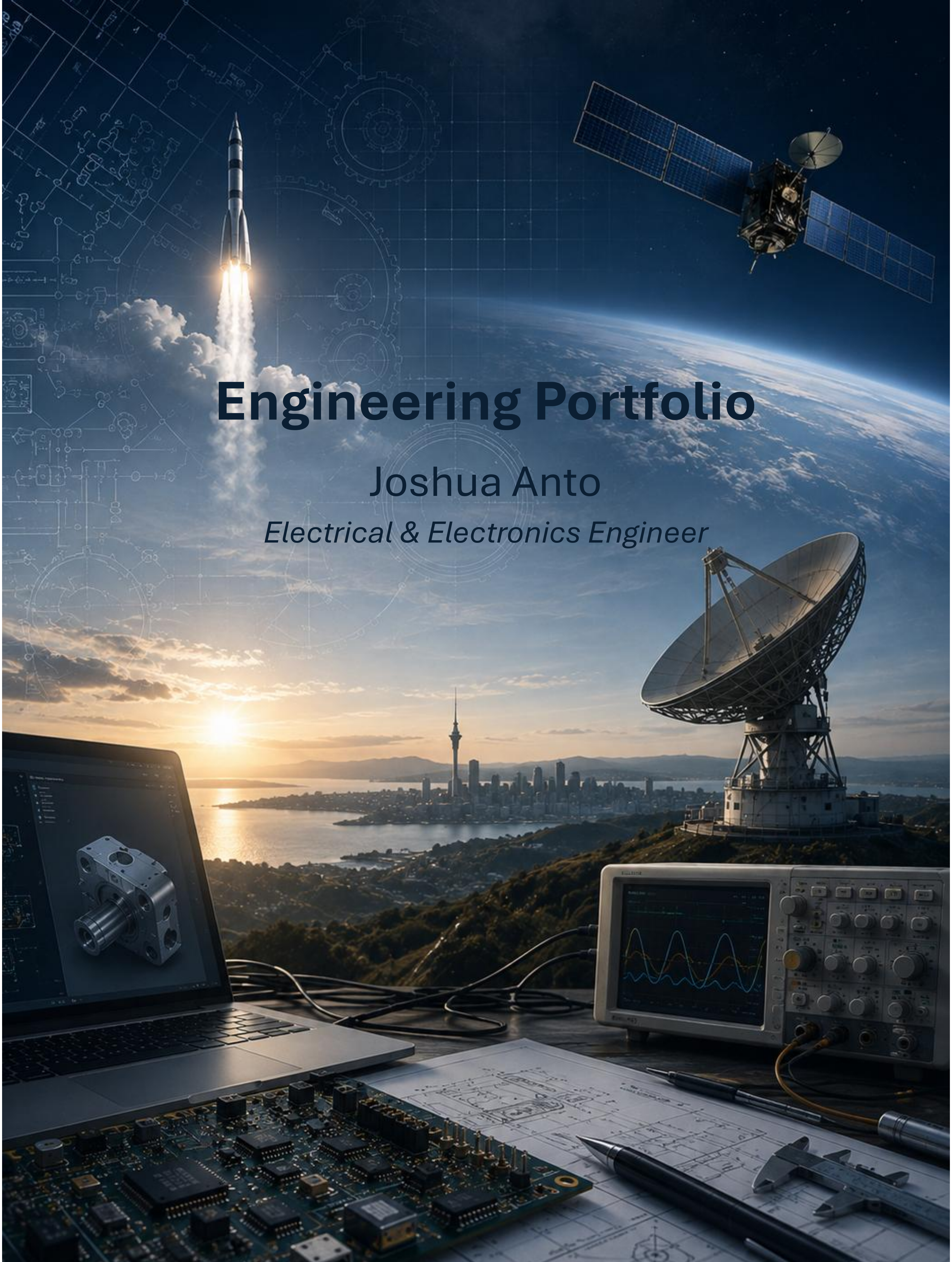

Engineering Portfolio

Joshua Anto

Electrical & Electronics Engineer

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About Me

I am an Electrical & Electronics Engineer and Master of Aerospace Engineering candidate at the University of Auckland with interests in spacecraft systems, embedded systems, RF communications, and system design. I enjoy tackling engineering problems that require balancing technical constraints and developing practical, well-reasoned solutions through systems thinking.

Throughout my studies and professional experience, I have worked on projects spanning embedded systems, PCB design, robotics, simulation, aerospace systems, and software development. These opportunities have strengthened both my technical knowledge and my ability to evaluate engineering trade-offs, integrate multidisciplinary systems, and approach complex problems with a structured mindset.

I value collaboration, continuous learning, and clear communication. Engineering competitions, international projects, and technical presentations have reinforced the importance of working effectively with others and communicating complex ideas in a way that is accessible to different audiences.

As I continue my Master's research, I hope to build a career in the space industry where I can contribute to the design, development, and operation of advanced aerospace systems. I am particularly motivated by projects that combine research, innovation, and systems engineering to solve challenging real-world problems and support the future of space exploration.

Skills Overview

Systems & Aerospace Engineering

- Systems Engineering
- Requirement Analysis
- System Architecture
- Trade Studies
- System Modelling
- Mission Analysis
- Interface Definition
- Link Budget Analysis
- Orbital Mechanics
- System Budgeting
- Subsystem Sizing

Electrical & Electronics Engineering

- Analogue Electronics
- Digital Electronics
- Electrical System Design
- Power Distribution
- Signal Flow Analysis
- RF Fundamentals
- Telecommunication Systems
- Electrical Troubleshooting

Embedded Systems & Programming

- Embedded Systems
- Hardware-Software Integration
- Embedded Development
- Embedded Debugging
- Microcontroller Programming
- Serial Communication
- Arduino
- Raspberry Pi
- Python
- C/C++
- Git/GitHub

Testing, Verification & Instrumentation

- System Testing
- Verification & Validation
- Functional Testing
- Functional Inspection
- Fault Finding
- Bench Diagnostics
- Electrical Measurements
- Test Equipment Operation
- Quality Inspection

CAD, PCB & Manufacturing

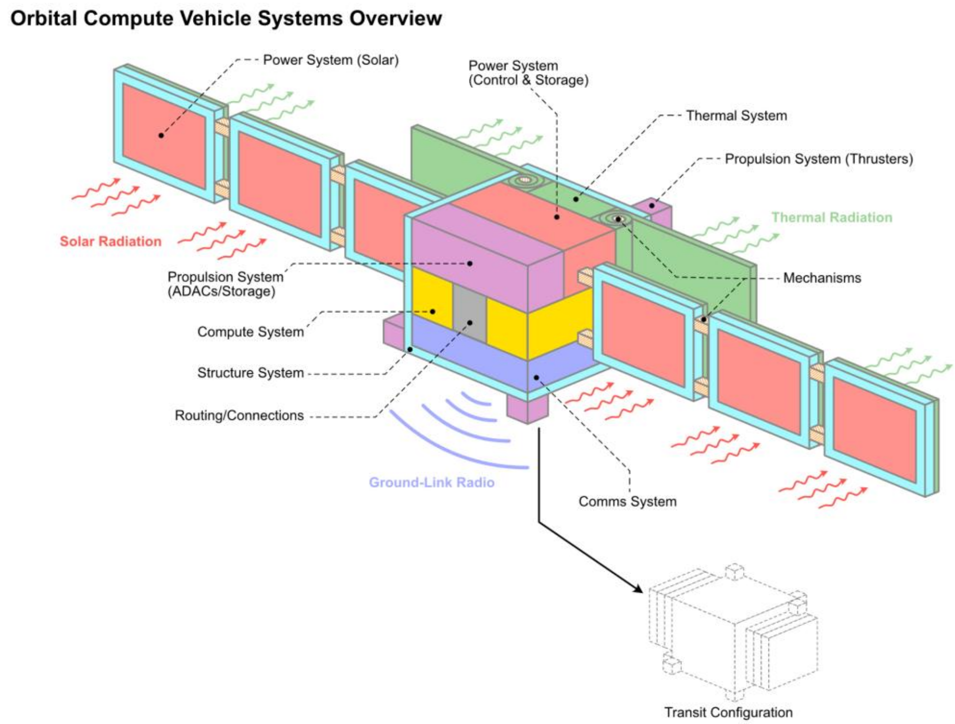
- Altium Designer
- SolidWorks
- Fusion 360
- PCB Design & Layout
- 3D Modelling
- Enclosure Design
- Wiring Harnesses Assembly
- Soldering & PCBA Rework
- Crimping & Cable Shielding
- Mechanical & Electrical Assembly

Simulation & Analysis

- MATLAB
- LabVIEW
- PowerCAD
- Multisim
- SUMO
- Engineering Simulations
- Data Analysis

Projects

Space-Based AI Data Centre System Concept



Project Summary

Project Type	University Systems Engineering Project
Status	Complete
Institution	University of Auckland
Discipline	Aerospace Systems Engineering
Role	Systems Engineering Team Member & Architect
Focus Areas	Mission Architecture, Systems Engineering, Programme Plan
Key Contribution	Led technical meetings, authored major systems engineering sections and modelled system.
Tools Used	• Google Docs • Google Sheets • Google Meet • Draw.io • Systems Engineering Methodologies • Technical Research
Duration	Semester 1, 2026
Team Size	8
Detailed Report	*View Full Report*

Overview

A multidisciplinary systems engineering project investigating the feasibility of a space-based AI data centre capable of supporting future computational demands while reducing reliance on terrestrial infrastructure. The project evaluated mission architecture, spacecraft systems, lifecycle planning, financial feasibility, and operational sustainability to produce a complete mission concept.

Objectives

- Research and develop a technically feasible concept for a space-based AI data centre.
- Design an end-to-end spacecraft system architecture.
- Evaluate operational, maintenance, and end-of-life strategies.
- Produce a realistic programme plan and financial assessment.
- Design and model system operations and interfaces
- Apply systems engineering principles throughout the project lifecycle.

Solution Overview

Following a series of systems engineering trade studies, our team selected a Hybrid LEO–GEO architecture as the preferred mission concept. The design separated AI model training and inference into different orbital segments to take advantage of the strengths of each orbit. A dedicated GEO platform was proposed for computationally intensive AI model training and large-scale data processing, as its continuous visibility to ground stations and stable orbital environment make it well suited for long-duration workloads and frequent data exchange.

A distributed constellation of LEO nodes was designed to provide low-latency AI inference services closer to Earth, improving responsiveness while allowing the system to scale through multiple interconnected satellites. Communication between the GEO platform and LEO constellation was achieved through a ground-mediated network that transferred training datasets, model updates, and inference data. This hybrid approach provided a practical balance between computational capability, operational efficiency, scalability, and long-term sustainability, making it the preferred solution over the alternative LEO-only, MEO, and GEO-only architectures considered during the project.

Engineering Process

The project followed a structured systems engineering methodology, using the Systems Engineering V-Model to guide development from mission definition through to the preliminary system design. Mission objectives, stakeholder requirements, and operational needs were progressively translated into system requirements, subsystem architectures, and verification planning, ensuring traceability throughout the design process. The preliminary design phase was managed using a Waterfall development approach, allowing each stage to build systematically on the completion of the previous one.

The project progressed through mission analysis, requirements definition, concept generation, architecture development, subsystem design, trade studies, lifecycle planning, and programme management before producing the final conceptual design. Trade studies evaluated alternative deployment, communication, and compute architectures against performance, cost, scalability, sustainability, and operational feasibility, ultimately leading to the selection of a Hybrid LEO–GEO architecture supported by a Protected COTS compute architecture as the preferred baseline solution. The final design demonstrated the application of systems engineering principles to a complex aerospace concept from mission conception through end-of-life planning. This led to achieving the highest grade in the cohort.

Skill Demonstrated & Contributions

I played a leading role in both the technical development and coordination of the project. My primary responsibilities included authoring and developing the following areas:

Mission Overview	Defined the mission purpose, objectives, and overall project scope.
Concept of Operations	Developed the operational concept describing how the system functions throughout the mission lifecycle.
Requirements Analysis	Translated stakeholder needs into system-level requirements to guide the design process.
System Overview	Produced a high-level overview of the proposed system and its major functional elements.
Engineering Trade Studies	Evaluated alternative architectures and design options to support engineering decision-making.
Systems Modelling	Modelled subsystem interactions and operational workflows within the proposed architecture.
System Architecture	Developed the Hybrid LEO–GEO architecture separating AI training and inference.
Maintenance Strategy	Developed the maintenance approach to support long-term operational reliability.
Upgrade Strategy	Planned system upgrade pathways to improve scalability and future adaptability.
Disposal Strategy	Defined the end-of-life disposal approach to support sustainable orbital operations.

Programme Plan	Produced the programme schedule, milestones, and implementation roadmap.
Life Cycle Planning	Developed the system lifecycle plan, defining activities from deployment and operation through maintenance, upgrades, and end-of-life disposal.
System Budgets	Developed engineering budgets for mass, power, communications, and other system resources to support the proposed mission architecture.
Financial Budget and Cost Planning	Prepared programme cost estimates and financial planning to assess project feasibility.
Lifecycle Engineering	Developed maintenance, upgrade, and disposal strategies for long-term sustainability.
Mission Planning	Defined mission objectives, deployment strategy, and operational roadmap.
Project Management	Produced project schedules, milestones, and implementation plans.
Cost & Budget Analysis	Estimated programme costs and assessed financial feasibility.
Technical Documentation	Authored system architecture, lifecycle, programme planning, and financial documentation.
Technical Leadership	Led technical meetings and facilitated engineering decision-making.
Team Collaboration	Worked within a multidisciplinary team to develop the final mission concept.

Challenges

One of the main challenges was balancing ambitious mission objectives with realistic engineering constraints. Designing a technically feasible system while considering maintenance, upgradeability, disposal, programme scheduling, and financial limitations required continuous collaboration and iterative design refinement. Leading technical discussions also involved integrating diverse viewpoints into a coherent systems solution.

Lessons Learned

This project strengthened my understanding of large-scale systems engineering by demonstrating how technical, operational, financial, and programme management considerations must all be integrated into a successful mission concept. It also developed my confidence in leading engineering discussions, coordinating multidisciplinary teams, and communicating complex technical ideas.

Images / Diagrams

In Development

Project JARVIS: AI Assistant

Project Summary

Project Type	Independent Engineering Project
Status	Prototype Development (Ongoing)
Institution	Independent
Discipline	Embedded Systems, Internet of Things, Artificial Intelligence
Role	Sole Developer
Focus Areas	Voice Recognition, Embedded Systems, Hardware–Software Integration, AI Integration, IoT
Key Contribution	Designed and developed the complete system architecture, implemented the Python–Arduino communication framework, integrated voice command functionality, and built the initial working prototype.
Tools Used	• Python • Arduino

	• Raspberry Pi • Visual Studio Code
Duration	2026 - Present
Team Size	Independent
Detailed Report	*In Development*

Overview

Inspired by J.A.R.V.I.S., the iconic virtual assistant from the *Iron Man* films, this independent engineering project aims to develop a modular voice-controlled assistant capable of interacting with embedded hardware. While fictional, J.A.R.V.I.S. has inspired engineers and technology enthusiasts by showcasing the potential of intelligent human–computer interaction, motivating the development of systems that bridge software and the physical world.

The long-term vision is to combine speech recognition, artificial intelligence, microcontrollers, and IoT technologies into a scalable platform capable of controlling real-world devices through natural language. The current prototype focuses on establishing a reliable communication framework between Python and an Arduino microcontroller, validating the core hardware–software architecture that will support future integration of voice recognition, large language models, computer vision, wireless connectivity, and intelligent automation.

Objectives

- Design and develop a modular AI-powered voice assistant capable of interacting with embedded hardware through natural language commands.
- Establish reliable communication between Python and Arduino to enable real-time control of physical devices.
- Develop a scalable system architecture that supports future integration of additional hardware, sensors, and intelligent software modules.
- Investigate the practical application of artificial intelligence, embedded systems, and IoT technologies within a unified engineering platform.
- Apply iterative prototyping and testing to validate hardware–software integration and improve system reliability.
- Build a long-term engineering platform that can evolve to incorporate advanced capabilities such as computer vision, large language models, autonomous decision-making, and smart automation.

Solution Overview

The current prototype establishes the core communication framework between a Python application and an Arduino microcontroller through serial communication. User commands entered the Python interface are transmitted to the Arduino, where they are interpreted and executed to control connected electronic components on a breadboard. This validates the underlying hardware–software communication pipeline that forms the foundation of the project.

At its current stage, the system successfully executes commands such as **"light on"**, **"light off"**, **"motor on"**, and **"motor off"**, allowing LEDs and a DC motor to be controlled through a simple command interface. The modular architecture has been designed to support future expansion, with planned developments including voice recognition, large language model integration, computer vision, Raspberry Pi deployment, wireless IoT connectivity, and support for additional sensors and actuators as the project evolves.

Engineering Process

The development of JARVIS has followed an iterative prototyping approach, with the initial objective of establishing reliable communication between software and embedded hardware before introducing more advanced functionality. The project began by defining the system requirements and selecting appropriate hardware and software platforms. An Arduino Uno was chosen as the embedded controller, while Python was selected as the primary interface for communicating with the microcontroller via serial communication over USB, providing a simple and reliable platform for validating the system architecture.

Following hardware assembly on a breadboard, Arduino firmware was developed to interpret predefined serial commands, while a Python application was implemented to transmit commands and control connected devices. Functional testing was carried out to verify reliable communication, command execution, and overall system stability using LEDs and a DC motor. The project continues to evolve through incremental development, with future iterations planned to incorporate voice recognition, computer vision, large language models, wireless connectivity, and broader IoT capabilities.

Skills Demonstrated & Contributions

As the sole developer of JARVIS, I have been responsible for the complete design, development, and testing of the prototype from concept through implementation. The project has provided an opportunity to apply embedded systems engineering principles while gaining practical experience in hardware–software integration.

My key responsibilities include:

System Concept & Architecture	Designed the overall system concept and a modular architecture to support future expansion.
Prototype Development	Designed and assembled the breadboard prototype, integrating LEDs, a DC motor, and supporting circuitry.
Embedded Systems Programming	Developed the Arduino firmware to receive and execute serial commands for hardware control.
Software Development	Developed the Python application to establish serial communication and transmit user commands.

Hardware–Software Integration	Integrating the hardware and software components into a functional prototype.
Testing & Debugging	Performed functional testing and debugging to validate reliable communication and device operation.
Future Development Planning	Defined the project roadmap for future integration of voice recognition, large language models, computer vision, and IoT connectivity.
Serial Communication	Implemented USB serial communication between the PC and Arduino.
Electronics Prototyping	Designed and assembled the breadboard prototype using LEDs, a DC motor, and supporting circuitry.
Circuit Integration	Integrated electronic components into a functional prototype for testing.

Challenges

One of the primary challenges was establishing reliable communication between the Python application and the Arduino microcontroller while ensuring commands were transmitted and executed consistently. Developing and debugging the serial communication interface required iterative testing to verify correct command interpretation, hardware response, and overall system stability before additional functionality could be introduced.

Another challenge was designing the prototype with future expansion in mind. Rather than developing a single-purpose controller, the system architecture was designed to remain modular so that features such as voice recognition, large language models, computer vision, and IoT connectivity can be integrated without requiring a complete redesign of the existing hardware–software framework.

Lessons Learned

Developing JARVIS reinforced the importance of building reliable foundations before introducing additional complexity. By focusing on hardware–software communication first, I gained a greater appreciation for validating individual subsystems through iterative prototyping before expanding overall system functionality.

The project also strengthened my understanding of embedded systems integration, modular software architecture, and the value of designing with future scalability in mind. It highlighted how thoughtful system design during the early stages can simplify future development and support the integration of more advanced capabilities as the project evolves.

Images / Diagrams

Content In Development

Aircraft & Atmosphere System Concept

Project Summary

Project Type	University Systems Engineering Project
Status	Complete
Institution	University of Auckland
Discipline	Aeronautical Systems Engineering
Role	Systems Engineering Team Member & Architect
Focus Areas	Aircraft Systems Design, Mission Analysis, Systems Engineering, Subsystem Integration
Key Contribution	Authored the Mission Overview, System Overview, subsystem architecture models, Electrical & Environmental Control System, and Avionics subsystem documentation.
Tools Used	• Google Docs • Google Sheets • Google Meet • Draw.io • Systems Engineering Methodologies • Technical Research
Duration	Semester 1, 2026
Team Size	6
Detailed Report	*View Full Report*

Overview

This university systems engineering project involved the conceptual design of a civilian maritime surveillance and search-and-rescue aircraft to complement New Zealand's existing P-8A Poseidon fleet. Working within operational, regulatory, and cost constraints, the project applied a structured systems engineering approach to develop a preliminary aircraft capable of conducting maritime surveillance, search and rescue, and aid delivery across New Zealand's extensive maritime area of interest.

The project progressed from mission definition and stakeholder analysis through requirements development, system architecture, subsystem integration, and preliminary aircraft design. Based on trade studies, the Pilatus PC-12 PRO Special Mission platform was selected as the preferred solution, integrating surveillance sensors, avionics, communication and navigation systems, and search-and-rescue capabilities into a practical, cost-effective aircraft concept that balanced operational performance, maintainability, and certification requirements.

Objectives

- Develop a conceptual civilian maritime surveillance aircraft to complement New Zealand's existing maritime patrol capability.
- Apply a structured systems engineering approach to define mission requirements, system architecture, and subsystem integration.
- Design an aircraft capable of conducting maritime surveillance, search and rescue, and emergency aid delivery while operating from regional airfields.
- Develop and integrate the aircraft's major subsystems, including avionics, electrical, environmental control, communications, and surveillance systems, into a cohesive preliminary design.
- Evaluate design decisions against operational performance, certification requirements, maintainability, lifecycle considerations, and programme cost to produce a practical and feasible mission solution.

Solution Overview

Following a series of systems engineering trade studies, the team selected the Pilatus PC-12 PRO Special Mission as the preferred platform for a civilian maritime surveillance and search-and-rescue aircraft. Rather than developing a new airframe, the solution leveraged an existing certified aircraft and integrated mission-specific capabilities, including maritime radar, EO/IR sensors, AIS reception, communications and navigation systems, mission management equipment, and emergency payload deployment. This approach reduced development and certification risk while delivering a practical, cost-effective capability for New Zealand's maritime operations.

The proposed system was designed to conduct long-range maritime surveillance, identify vessels during day and night operations, coordinate with rescue agencies, and deploy life-saving equipment when required. The PC-12 was selected because it offered the best balance of range, short-field performance, operating cost, maintainability, and mission-system integration compared with alternative aircraft considered during the trade studies, providing a scalable solution that complements, rather than replaces the RNZAF P-8A Poseidon fleet.

Engineering Process

The project followed a structured systems engineering approach, progressing from mission definition and stakeholder analysis through requirements development, concept generation, trade studies, subsystem design, and verification planning. Mission objectives were translated into system-level requirements before being decomposed into functional architectures and integrated aircraft subsystems, ensuring traceability throughout the preliminary design process.

Trade studies were undertaken to evaluate candidate aircraft platforms against operational capability, cost, maintainability, certification requirements, and mission suitability, leading to the selection of the Pilatus PC-12 PRO Special Mission as the preferred baseline. The preliminary design was then refined through subsystem architecture development, interface modelling, and verification planning to ensure the proposed aircraft satisfied the defined mission and operational requirements.

Skills Demonstrated & Contributions

This project strengthened my understanding of aircraft systems engineering, requirements development, subsystem integration, and systems architecture.

Mission Overview	Developed the mission overview, including project scope, objectives, stakeholders, ConOps, and success criteria.
Requirements Analysis	Defined the top-level system requirements from mission and stakeholder needs.
System Overview	Produced the overall system overview and high-level aircraft architecture.
Subsystem Architectures & Interface Modelling	Created all subsystem architecture and interface diagrams illustrating system integration.
Electrical & Environmental Control System	Designed and documented the Electrical and Environmental Control System.
Avionics Subsystem	Designed and documented the avionics subsystem, including navigation, communication, and surveillance systems.
Systems Engineering	Applied a structured systems engineering process from mission definition to preliminary aircraft design.
Mission Analysis Requirements Analysis	Defined mission objectives, operational needs, and success criteria. Translated mission and stakeholder needs into system-level requirements.

Concept of Operations Design	Developed the operational concept describing how the aircraft would fulfil its mission.
System Architecture	Produced the high-level aircraft architecture and subsystem integration.
Subsystem Design	Designed the Electrical & Environmental Control System and Avionics subsystem.
Systems Modelling	Created subsystem architecture and interface models to support system integration.
Interface Engineering	Defined subsystem interfaces to ensure functional integration across the aircraft.
Engineering Trade Studies	Evaluated design decisions against operational, certification, cost, and maintainability requirements.
Aircraft Systems Engineering	Applied systems engineering principles to the conceptual design of a special mission aircraft.
Technical Documentation	Authored key systems engineering sections of the preliminary design report.
Team Collaboration	Collaborated within a multidisciplinary team to develop an integrated aircraft concept.

Challenges

One of the primary challenges was balancing mission capability with operational, certification, and cost constraints while selecting a suitable aircraft platform. Multiple design trade-offs were required to ensure the proposed solution satisfied the mission requirements without introducing unnecessary complexity or development risk.

Another challenge was integrating multiple aircraft subsystems into a cohesive architecture while maintaining clear interfaces and traceability between mission objectives, system requirements, and subsystem designs. This required close collaboration across the team to ensure all engineering decisions supported the overall mission concept.

Lessons Learned

This project strengthened my understanding of how systems engineering connects mission needs to an integrated aircraft design. I learned the importance of defining clear requirements and interfaces early, as decisions within one subsystem can affect performance and integration across the entire aircraft.

It also highlighted the value of trade studies, multidisciplinary collaboration, and design traceability when balancing competing factors such as mission capability, cost, certification, maintainability, and operational constraints. Developing the subsystem architecture and interface models gave me a stronger appreciation for viewing an aircraft as an interconnected system rather than a collection of individual components.

Images / Diagrams

Content in Development

Orbital Propagation & Mission Design

Content In Development

Supporting Material

View Full Report

Smart Transportation System

Content In Development

Supporting Material

View Full Report

Wireless Power Transfer

Content In Development

Supporting Material

View Full Report

Embedded System Design

Content In Development

Supporting Material

View Full Report

Autonomous Navigation Robot

Content In Development

Supporting Material

View Schematic & PCB Design

Research

Orbital Utility Hub

Status: Independent Research (Ongoing)

The Orbital Power Hub is an independent research initiative exploring a modular orbital infrastructure capable of providing supplemental power, communications, and mission support services to spacecraft in Earth orbit. The concept investigates how shared space-based utilities could improve mission flexibility, reduce spacecraft design constraints, and support the long-term growth of the space economy.

Current research focuses on mission architecture, orbital trade studies, systems engineering, power transmission methods, communications, and operational feasibility. The project represents a long-term research interest that I intend to further develop through postgraduate research and future industry collaboration.

Achievements

National Finalist – Falling Walls Lab Aotearoa New Zealand

2026 | Falling Walls Foundation New Zealand

Selected as one of the national finalists from across New Zealand to present an original aerospace innovation at Falling Walls Lab Aotearoa New Zealand. Presented the Orbital Utility Hub, a concept proposing orbital infrastructure to enhance spacecraft power and mission capabilities through systems engineering.

Winning Team – TUC Space Summer School Innovation Challenge

2026 | Technical University of Crete

Awarded 1st place after collaborating with an international team to develop and present an innovative space mission concept during the inaugural TUC Space Summer School. The challenge required teams to devise a feasible response to a hypothetical asteroid on a collision course with Earth, with only one month before impact. Our solution combined mission design, systems thinking, rapid decision-making, and effective teamwork to deliver the winning proposal among a competitive international cohort.

New Zealand & Oceania Representative – TUC Space Summer School

2026 | Technical University of Crete

Selected as the only participant representing New Zealand and the Oceania region at the inaugural TUC Space Summer School in Greece, joining an international cohort to explore spacecraft systems, mission design, space technologies, and astronaut selection.

Leadership, Communication & Outreach

International Engineering Collaboration

2026 | TUC Space Summer School – Greece

Collaborated with an international, multidisciplinary team to develop an innovative space mission concept under strict time constraints. Worked closely with team members from diverse backgrounds to evaluate ideas, integrate technical solutions, and deliver a successful final presentation.

Entrepreneurial Innovation Pitch

2026 | Falling Walls Lab Aotearoa New Zealand

Presented the Orbital Utility Hub concept as a national finalist at Falling Walls Lab Aotearoa New Zealand. Delivered a three-minute presentation explaining a complex aerospace systems concept to judges and a multidisciplinary audience, demonstrating concise technical communication under strict time constraints.

Technical Innovation Presentation

2026 | ActInSpace New Zealand – SpaceBase Limited

Worked within a multidisciplinary team to develop and pitch an innovative solution to a real-world space challenge. Combined engineering problem-solving with entrepreneurial thinking by communicating both the technical feasibility and commercial potential of the proposed concept. Supported by CNES and the European Space Agency.

Engineering Project Exhibition

2024 | Auckland University of Technology

Presented my final-year engineering project of a smart transportation system, during the university's Engineering Project Exhibition. Demonstrating the design, development, and outcomes of the project to academic staff, industry representatives, and visitors. Answered technical questions and communicated engineering decisions to audiences with varying levels of technical knowledge.

Business & Systems Concept Presentation

2022 | Auckland University of Technology

Developed and presented a comprehensive Business Model Canvas for SECUREKIWI NZ LTD, a conceptual marine fire alarm company. The project required integrating engineering design with commercial strategy by defining the value proposition, customer segments, partnerships, revenue streams, and operational model, while presenting the concept to academic staff and judges

Muay Thai Club Event Coordinator

2021 | Auckland University Muay Thai Club

Coordinated club events by organising logistics, scheduling activities, and communicating with members to ensure successful event delivery. The role strengthened my leadership, teamwork, organisational, and interpersonal skills while supporting an active university community.

Contact Information

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GitHub: https://github.com/Joshuaanto

Website: *In Development*

Projects Archive

Project: Cosmological Distance & Supernova

Discipline: Astrophysics

Description: A computational astrophysics project involving numerical cosmology modelling and analysis of supernova datasets to study luminosity distances and expanding-universe behaviour.

Supporting Material: *In Development*

Project: Home Security System

Discipline: Embedded Software Engineering

Description: An embedded software engineering project using a Raspberry Pi and Sensor Hat to create a smart home security and monitoring system with remote database integration.

Supporting Material: *View Full Report*

Project: Embedded System Utilizing Computer Vision & ML

Discipline: Embedded Software Engineering

Description: A computer vision and machine learning project using OpenCV, TensorFlow, and Raspberry Pi to create a gesture-recognition Rock-Paper-Scissors game

Supporting Material: *View Full Report*

Project: Analog & Digital Piano

Discipline: Electronics Engineering

Description: A PCB and embedded systems project involving analogue and digital PWM audio generation using NE555 timers and ATtiny microcontrollers.

Supporting Material: *View Full Report*

Project: Dual Rail Power Supply

Discipline: Electronics Engineering

Description: A PCB design and electronics project involving the development of a half-wave bridge dual-rail power supply with regulated voltage outputs.

Supporting Material: *View Full Report*

Project: Mobile Solar Powered Trailer

Discipline: Engineering Design

Description: A systems engineering and CAD project involving the design of a mobile solar-powered bore pump trailer for remote communities in Cape York.

Supporting Material: *View Full Report*